Notes on the proof of the remainder

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Assumption 0.1 (Trait-direction stability estimates). *On the time interval or for the stationary state under consideration, assume that the following estimates hold.*

(B1) The discrete phase satisfies

$$\max_k \left(|D_\theta^- u_k^\varepsilon| + |D_\theta^+ u_k^\varepsilon|\right) \le L_T.$$

(B2) The amplitude satisfies the weighted first-difference estimate

$$\Delta\theta \sum_{k=1}^{K} \left(E_k^\varepsilon + E_{k+1}^\varepsilon\right) |D_\theta^+ W_{j,k}^\varepsilon| \le M_T m_j^\varepsilon.$$

(B3) The numerical Hamiltonian is bounded on bounded slopes,

$$\left|\widehat{H}\left(D_\theta^- u_k^\varepsilon, D_\theta^+ u_k^\varepsilon\right)\right| \le C_H L_T^2.$$

(B4) The conservative trait flux satisfies the weighted interface bound

$$E_k^\varepsilon \left|\widehat{\mathcal{F}}_{j,k+1/2}\right| \le C_F L_T \left(n_{j,k}^\varepsilon + n_{j,k+1}^\varepsilon\right),$$

and the analogous estimate holds for the interface $k - 1/2$.

Remark 0.1. *For the Godunov Hamiltonian $\widehat{H}_G$ and the upwind flux given above, (B3) and (B4) hold on bounded slope sets. Indeed,*

$$|\widehat{H}_G(p^-, p^+)| \le |p^-|^2 + |p^+|^2,$$

and the upwind choice avoids uncontrolled ratios of neighboring exponential weights.

Lemma 0.1 (One-sided control of the weighted trait remainder). *Assume that Assumption* **??** *holds. Then*

$$\mathcal{R}_j^{\varepsilon,D} \le C\left(L_T^2 + \frac{\varepsilon L_T}{\Delta\theta} + \frac{\varepsilon^2 M_T}{\Delta\theta}\right) m_j^\varepsilon. \tag{0.1}$$

In particular, for fixed $\Delta\theta > 0$ and $0 < \varepsilon \le 1$, there exists a constant $C_R = C_R(\Delta\theta, L_T, M_T)$, independent of ε, such that

$$\mathcal{R}_j^{\varepsilon,D} \le C_R m_j^\varepsilon. \tag{0.2}$$

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Proof. We estimate the four terms in $R^{\varepsilon}_{j,k}$. Throughout the proof, C denotes a constant independent of ε.

For the first term, using

$$\delta^2_\theta W^{\varepsilon}_{j,k} = \frac{D^+_\theta W^{\varepsilon}_{j,k} - D^+_\theta W^{\varepsilon}_{j,k-1}}{\Delta\theta},$$

we obtain

$$\begin{aligned}
&\varepsilon^2 \Delta\theta \sum_{k=1}^{K} \frac{E^{\varepsilon}_k}{D_k} \left|\delta^2_\theta W^{\varepsilon}_{j,k}\right| \\
&\quad\le C\frac{\varepsilon^2}{\Delta\theta}\Delta\theta \sum_{k=1}^{K} \left(E^{\varepsilon}_k + E^{\varepsilon}_{k+1}\right)\left|D^+_\theta W^{\varepsilon}_{j,k}\right| \\
&\quad\le C\frac{\varepsilon^2 M_T}{\Delta\theta} m^{\varepsilon}_j.
\end{aligned}$$

For the second term, (B1) gives

$$|\delta^2_\theta u^{\varepsilon}_k| = \left|\frac{D^+_\theta u^{\varepsilon}_k - D^+_\theta u^{\varepsilon}_{k-1}}{\Delta\theta}\right| \le \frac{2L_T}{\Delta\theta}.$$

Therefore,

$$\begin{aligned}
&\varepsilon\Delta\theta \sum_{k=1}^{K} \frac{n^{\varepsilon}_{j,k}}{D_k} \left|\delta^2_\theta u^{\varepsilon}_k\right| \\
&\quad\le C\frac{\varepsilon L_T}{\Delta\theta} m^{\varepsilon}_j.
\end{aligned}$$

For the Hamiltonian term, (B3) yields

$$\begin{aligned}
&\Delta\theta \sum_{k=1}^{K} \frac{n^{\varepsilon}_{j,k}}{D_k} \left|\widehat{H}\left(D^-_\theta u^{\varepsilon}_k, D^+_\theta u^{\varepsilon}_k\right)\right| \\
&\quad\le C L_T^2 m^{\varepsilon}_j.
\end{aligned}$$

Finally, for the flux term,

$$\begin{aligned}
&2\varepsilon\Delta\theta \sum_{k=1}^{K} \frac{E^{\varepsilon}_k}{D_k} \left|\mathcal{D}_\theta \widehat{\mathcal{F}}_{j,k}\right| \\
&\quad\le C\varepsilon \sum_{k=1}^{K} E^{\varepsilon}_k \left(|\widehat{\mathcal{F}}_{j,k+1/2}| + |\widehat{\mathcal{F}}_{j,k-1/2}|\right) \\
&\quad\le C\frac{\varepsilon L_T}{\Delta\theta} m^{\varepsilon}_j.
\end{aligned}$$

Combining the four estimates gives (**??**). The bound (**??**) follows for fixed $\Delta\theta$ and $0 < \varepsilon \le 1$. $\square$